

MADEO: Modeler & Analyzer for Direct-wind Electromagnetic Optimization

Shane O'Brien¹, Febin Kurian², Ramesh Gupta²

¹Dept. of Physics & Astronomy, University of California, Irvine, CA, USA, 92697

²Superconducting Magnet Division, Brookhaven National Laboratory, Upton, NY, USA, 11973

Abstract:

The Superconducting Magnet Division (SMD) at Brookhaven National Laboratory (BNL) has the unique capability of developing direct wind magnets, like those that will be used in the Electron-Ion Collider (EIC). This study is the second stage in the development of a new program to model and find optimal configurations for direct wind magnet construction through a new Python based program, complete with a furnished graphical user interface (GUI). Development of a command-line interface version of this program was completed by hand, and then large-language models (LLMs) were used to aid in the development of the GUI and to make the optimization process run much faster than the original version. Through this new program, we were able to effectively find optimal conductor placement for different example magnet configurations. In one example, (). This new program, MADEO, will allow magnet designers at the SMD at BNL to design magnets in a faster and more user-friendly way with better visualizations of the actual magnet designs. This study aligns with Brookhaven's mission to support the Department of Energy's goal of developing higher quality superconducting magnets for future experiments, such as the EIC.

Introduction: Direct-Wind Magnets:

- Direct-wind magnets are superconducting coils wrapped around a support structure, such as an accelerator beam tube (Fig. 1). Because they can carry very high currents, they produce strong magnetic fields, and by changing the coil geometry, those fields can be shaped for different experimental needs.
- In particle accelerators, the strength and type of magnetic field matter because they directly affect how particles move. For dipole magnets, the field strength determines how strongly the beam is bent and therefore links the accelerator's bending radius to the beam energy it can reach (Fig. 2).
- A stronger, cleaner dipole field helps an accelerator reach higher energies with less complexity and energy cost. But dipoles are only one type of accelerator magnet. Quadrupole magnets focus the beam by pushing particles toward the center of the beamline, while sextupole magnets help keep particles with different energies properly focused.



(Fig. 1: A direct-wind magnet being wound at BNL [1])

$$F_{\text{Mag}} = ma_c \implies qvB = m \frac{v^2}{r} \implies qvB = (mv) \frac{v}{r} \implies \boxed{p = qBr}$$
$$E \approx pc \implies \boxed{E \approx (qBr)c}$$

(Fig. 2: The equation that relates the energy of a particle in a loop to the dipole field applied to it)

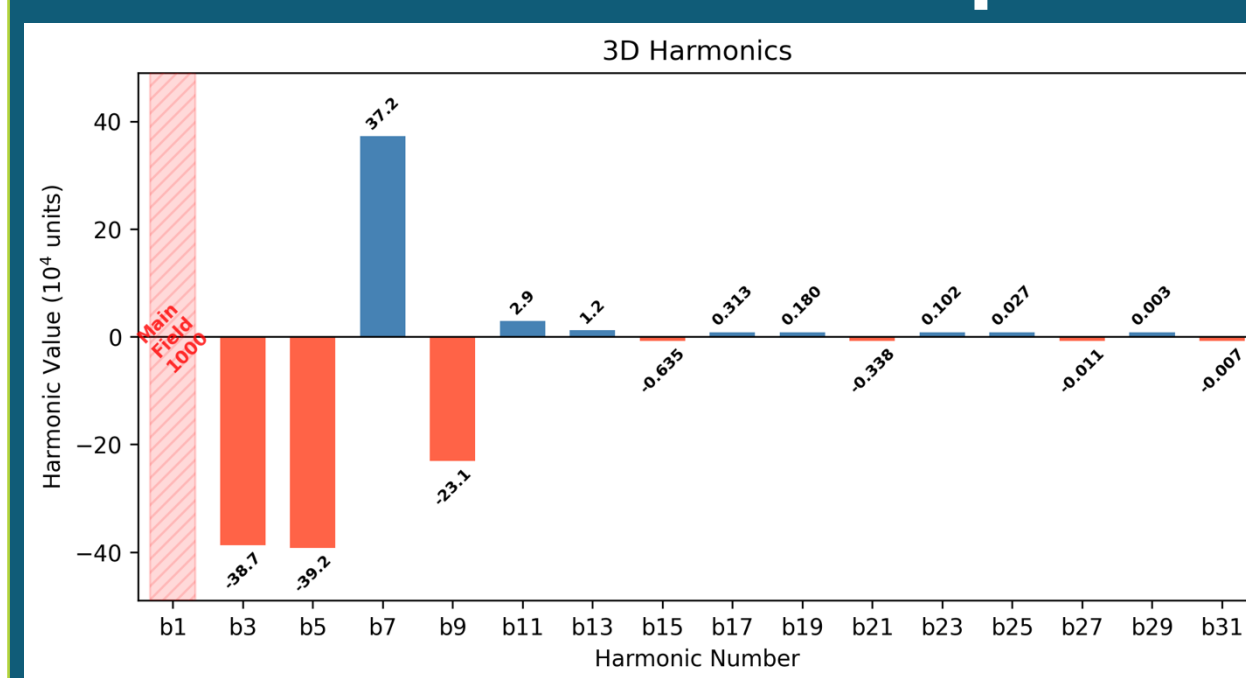
Introduction: Field Harmonics:

- Magnetic fields are composed of an infinite sum of terms called *harmonics* and *skew harmonics*, the mathematical definitions of which are shown in Fig. 3. Each type of magnet, or *pole*, has an associated harmonic term
- To ensure the desired accelerator field, or *main field*, is maximized, we want to minimize undesired field harmonics, so that the main field has the best quality it can have. The region through which we want to minimize these undesired harmonic terms is called the *reference radius*. This is the main radius through which the beam passes in an accelerator
- MADEO is a program that minimizes these undesired harmonics by finding the optimal spacing between conductor blocks

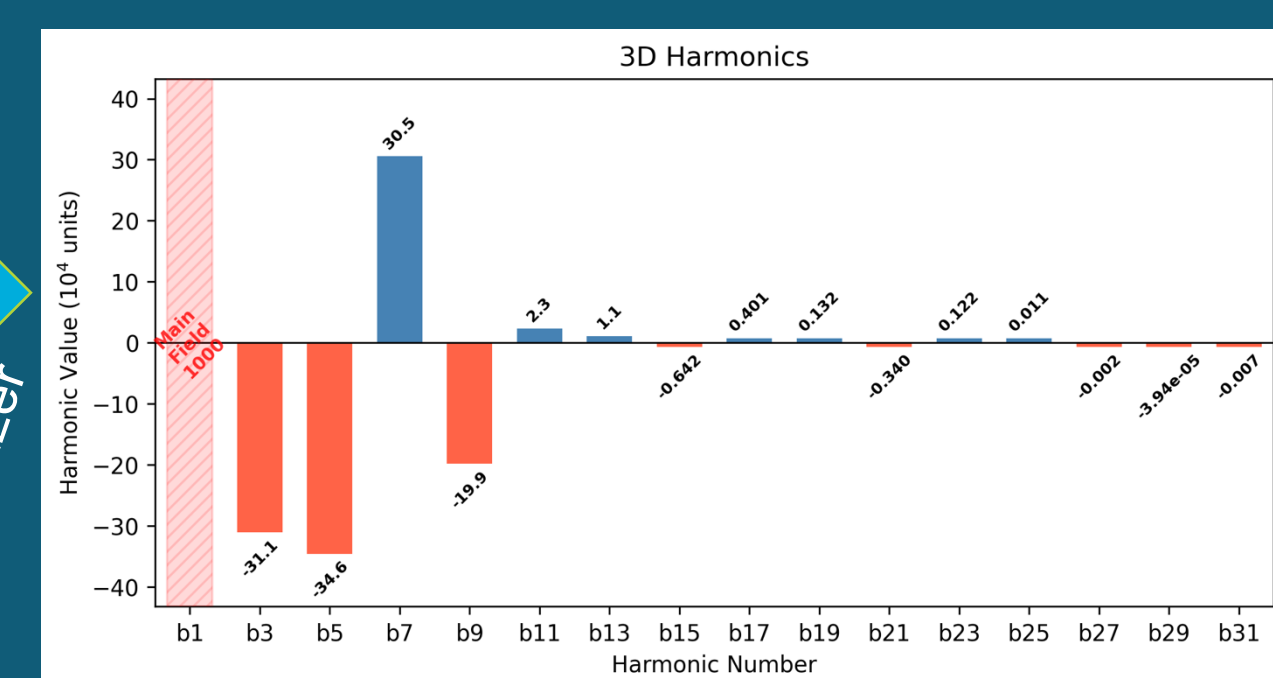
$$b_n = \frac{\mu_0 I}{2\pi} \left(\frac{R_0}{a} \right)^n \cos[(n+1)\phi]$$
$$b_{\text{norm}} = \frac{b_n}{b_{\text{main}}} \cdot 10^4$$
$$a_n = -\frac{\mu_0 I}{2\pi} \left(\frac{R_0}{a} \right)^n \sin[(n+1)\phi]$$
$$a_{\text{norm}} = \frac{a_n}{b_{\text{main}}} \cdot 10^4$$

(Fig. 3: The equations for the harmonics (b) as well as the skew harmonics (a) and their normalized form [2])

Results: Undesired Harmonic Reduction & Improved Visualization:



(Fig. 6: Harmonics pre-optimization)

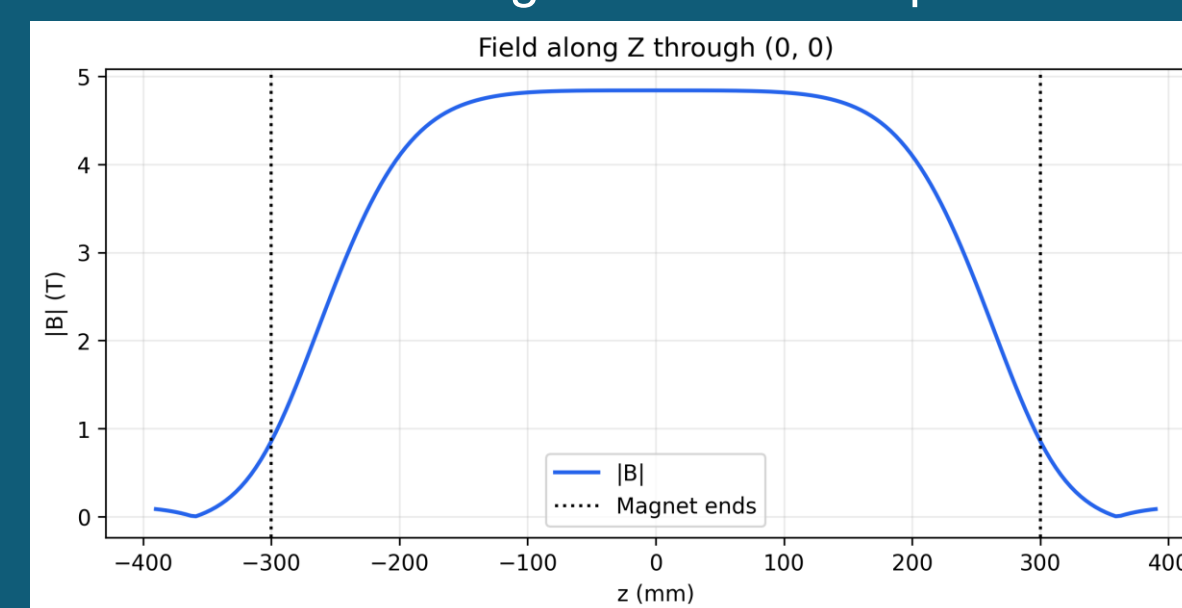


(Fig. 7: Harmonics post-optimization)

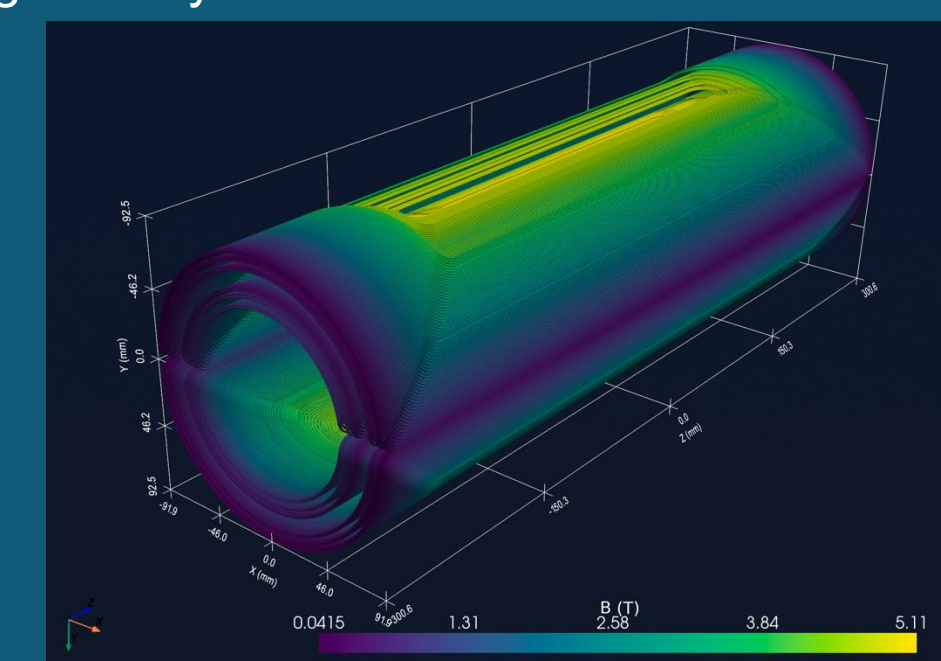
Harmonic	Weight	Pre-Optimization	Post-Optimization
b_1	0	10000.0000	10000.0000
b_3	1	-38.6835	-31.0763
b_5	1	-39.1783	-34.5838
b_7	2	37.2352	30.5475
b_9	3	-23.1072	-19.8500
b_{11}	4	2.9237	2.3125

(Table 1: Pre & post-optimization harmonic values for a 2-layer magnet example)

- In the above example, we take a 2-layer magnet and run it through the MADEO optimization algorithm
- We can see that the values produced by MADEO are indeed reduced in accordance with their individual weights
- Notably, these results also match the optimization results of the legacy Fortran code as well, confirming that MADEO is coming to the correct optimal values and geometry



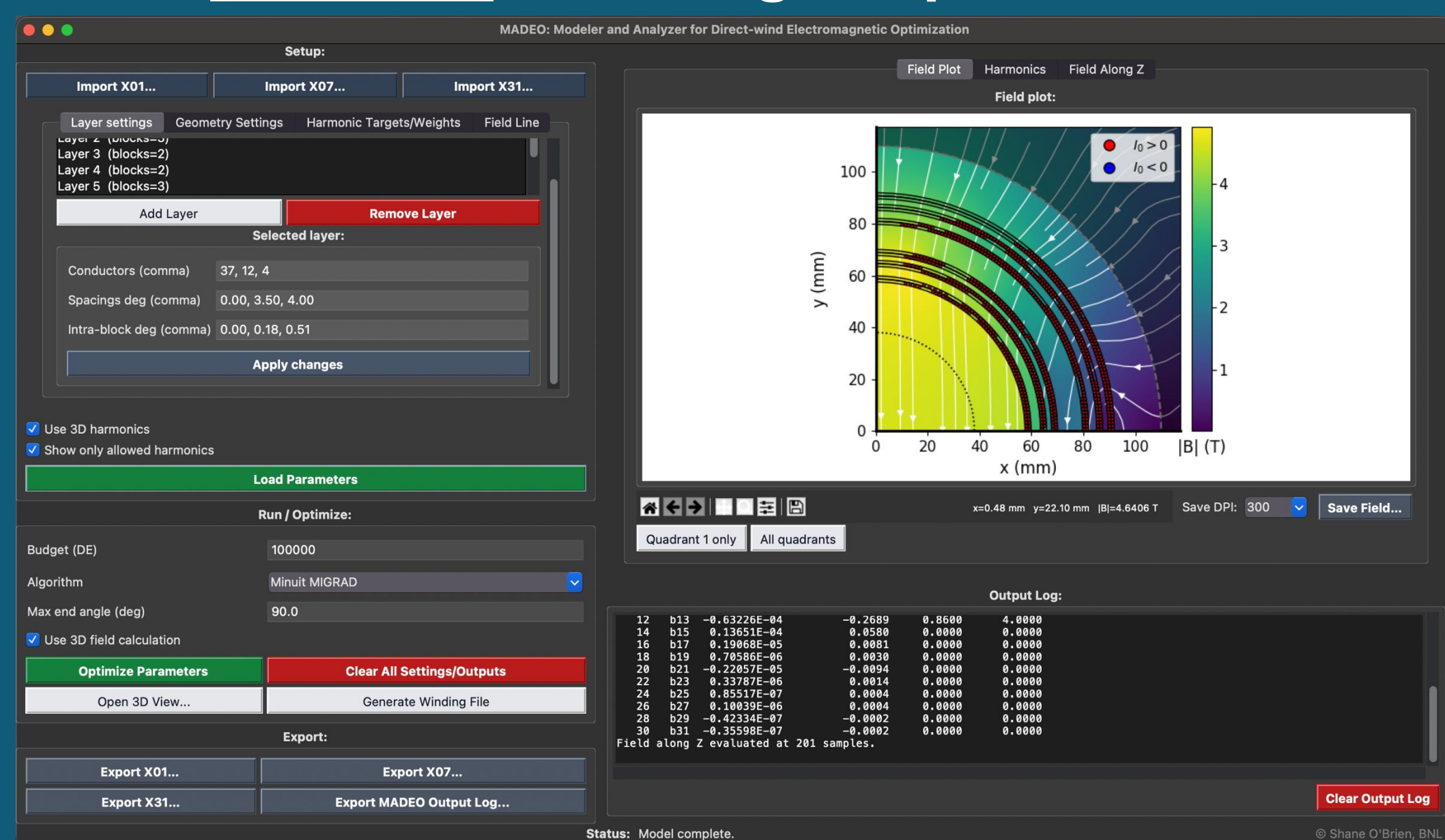
(Fig. 8: Field value through a magnet in 3D)



(Fig. 9: 3D model of a 12-layer magnet)

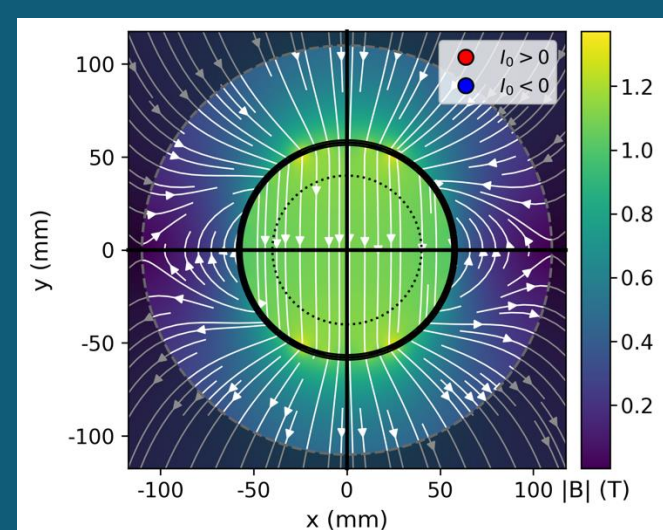
- MADEO also builds upon existing practices, and allows magnet designers to perform complex calculations and visualizations all within the same application, like above, where we see the field along the Z=0 line and a 3D model, both for a 12-layer magnet design
- It also allows magnet designers to export their final, optimized product to a winding file, allowing designers to move more efficiently from design to actual production

Methods: Modeling & Optimization:

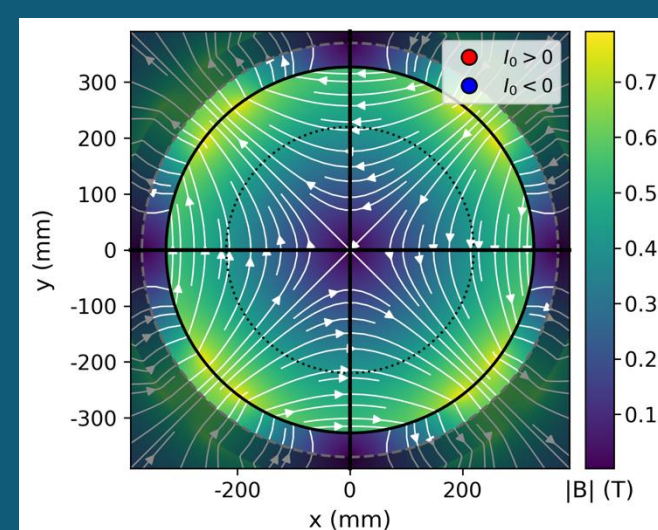


(Fig. 4: The GUI interface for MADEO that a user would see on start-up)

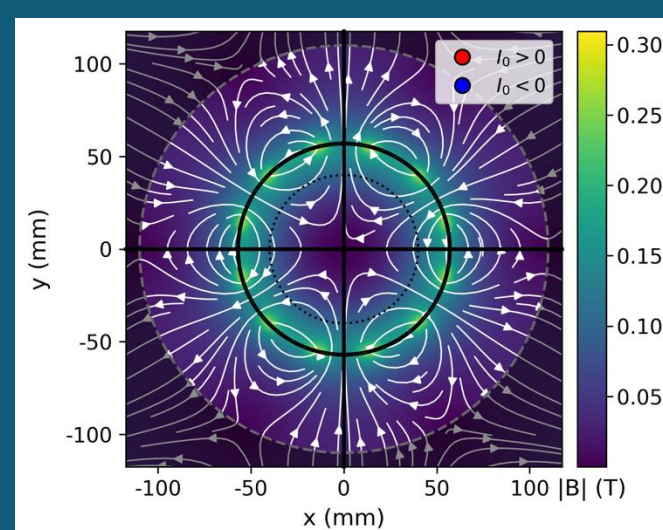
- To use MADEO, the user gives a collection of settings, from which an initial magnet design can be modeled. This done via config files or through the user manually inputting each setting themselves.
- These direct-wind magnets are modeled by plotting conductors in an arrangement provided by the user settings. Calculations are then done, of both the individual harmonics and the total magnetic field through the region
- The user can then run this preliminary design through an optimizer, that will then vary geometric parameters for each block in each layer until a configuration with an optimal field quality is found
- In order to optimize these parameters, we use the iMINUIT Python packages, which allows us to find these minimal values very quickly



(Fig. 5a: A dipole magnet modeled with MADEO)



(Fig. 5b: A quadrupole magnet modeled with MADEO)



(Fig. 5c: A sextupole magnet modeled with MADEO)

Discussion:

Although the program produced promising results, the final version of the code does not attempt to vary the conductor count within each block. In future work, this capability should be added so that the optimization can further reduce unwanted harmonic terms. In addition, the current version of the code does not account for the iron yoke that is typically placed around the magnet core to help guide the field lines back toward the center of the magnet. Including this feature in a later version would allow the model to better represent the true geometry and field behavior of these magnets. One final addition that would need to be included in a final version as well is considering the insulation of the wire. The wire insulation would impact the harmonics calculated as it would reduce the wire radius.

Acknowledgements:

A special thank you to my mentor Febin Kurian, as well as Ramesh Gupta for their support and guidance through this process. I would also like to thank Brookhaven National Lab and the Office of Educational Programs for the opportunity to pursue this work. This project was supported in part by the U.S. Department of Energy, Office of Science, Office of Workforce Development for Teachers and Scientists (WDTS), under the Science Undergraduate Laboratory Internships Program (SULI).

NO EXPORT CONTROL

Conclusion:

Through this project, we were able to successfully develop MADEO: a more modern, Python-based program for modeling and optimizing direct wind magnets with an intuitive and easy to use GUI. In addition to developing an algorithm to model and optimize the conductor configurations and the fields they produce, we also laid the groundwork for a more user-friendly replacement to the existing legacy Fortran code currently used to optimize magnet designs. Further adoption of MADEO will allow magnet designers at Brookhaven National Lab to be able to simulate magnets much quicker and not have to worry about editing any sort of files or changing any code themselves. Most importantly, it will give the magnet designers the ability to see their designs and optimization in real time.

References:

- [1] Brookhaven National Laboratory, "Initial Winding of BEPCII Magnets," BNL, <https://www.bnl.gov/magnets/pepii/initial-winding.php>.
- [2] Gupta, R. *Improving the Design and Analysis of Superconducting Magnets for Particle Accelerators*. Brookhaven National Laboratory, 1996, <https://wpw.bnl.gov/rgupta/wp-content/uploads/sites/9/2023/02/thesisrg.pdf>.